Towards Trustworthy IoT Network Security: Interpretable Open-Set Intrusion Detection

Ngari Crisphine Macharia and Ning Weng

***Abstract*—Machine learning can detect attacks in Internet of Things (IoT) networks, but operators cannot see why a model raises an alert, and new attacks are mistaken for known ones. We test whether interpretable models solve both. We compare Concept Bottleneck Models (CBMs), which classify through human-defined concepts, with NeSy-NIDS, which learns thresholds for expert-written rules alongside a neural network. On CTU-IoT-23 and CIC-IoT-2023, both models classify known traffic as accurately as a neural network. On later CTU-IoT-23 traffic every model misses one attack class, and on held-out CTU-IoT-23 source addresses the learned models' mean F1 falls to 0.56 or below, unlike on TON_IoT. Explanations only partly account for decisions: giving a CBM the declared concept values instead of its learned ones lowers F1 from 0.933 to below 0.50, and NeSy-NIDS's class predictions agree with its neural part on 97–100% of flows but with its rules on only 32–55%. For unfamiliar attacks, pulling concepts closer to their definitions has no consistent effect, the best detection score depends on dataset and model, and two ways of choosing it from familiar traffic each failed on a dataset. A CBM that learns concepts before its classifier loses accuracy but detects unfamiliar CTU-IoT-23 attacks about as well as the network. Seven measures of how well familiar classes separate track detection on CIC-IoT-2023 and TON_IoT, but on CTU-IoT-23 one model configuration drives the trend, and none predicts detection consistently on an unseen dataset. Interpretable intrusion detectors should be evaluated across several training runs, models and datasets.**

***Index Terms*—Concept bottleneck models, Internet of Things, network intrusion detection, open-set recognition, out-of-distribution detection.**

I. Introduction

Internet of Things (IoT) devices make up a large share of network endpoints, and their irregular updates and heterogeneous firmware make them attractive targets for botnet recruitment, DDoS amplification, and command-and-control (C&C) infrastructure. Because IoT devices cannot run endpoint agents, detection has to work from flows captured at a gateway or router tap [1]. The attack mix has recognizable signatures: Mirai-family botnets send high-rate UDP and TCP floods [2], C&C heartbeats beacon at regular inter-arrival times, and reconnaissance scans produce asymmetric packet counts. Such signatures lend themselves to rule- and concept-based representations. At the same time, new Mirai sub-families and DDoS variants keep appearing, so a model trained on a handful of attack families will encounter structurally similar but unseen variants in deployment.

N. C. Macharia and N. Weng are with the School of Electrical, Computer and Biomedical Engineering, Southern Illinois University Carbondale, IL 62901 USA (e-mail: c.ngari@siu.edu, nweng@siu.edu). Corresponding author: N. C. Macharia.

Code, data splits, pre-trained models, saved results and a script that maps them to every table are available at https://github.com/crisphinen/interpretable-nids.

Deep-learning NIDS classify known attacks accurately but raise two problems in deployment. The first is *opacity*: without knowing which traffic features drove a decision, operators cannot validate alerts, adjust policies, or meet regulatory requirements [3], [4]. Post-hoc explainers such as xNIDS [5] give more faithful explanations than LIME or SHAP, but they still explain a black box after the fact. The second is the *open-set problem*: a NIDS trained on a closed set of attacks meets new families in deployment [4] and confidently assigns them to known classes, and common remedies such as retraining or a separate anomaly detector do not use the model's own representation for open-set scoring [6].

In this paper, we study both problems together by building interpretability into the architecture and measuring what that does to open-set detection. In **CBMs** [7], predictions pass through a layer of human-defined flow concepts that can be inspected and overridden at test time. In **NeSy-NIDS**, expert attack signatures become threshold rules whose thresholds are learned through k-annealed sigmoid gates; a learned weight blends the rule head with a neural fallback, and a binarized rule vector is kept for audit and OOD scoring.

Contributions.

1) **Accuracy without full fidelity.** Both interpretable architectures match an unconstrained MLP on known classes from source addresses seen in training, but their declared concepts and rules only partly account for their class predictions; on held-out CTU source addresses the learned models fail to classify known traffic, while on TON_IoT classification is partly retained.
2) **Where interpretable representations help detection.** NeSy-NIDS gives the highest CTU OOD AUROC ($+0.030$ over the MLP, 95% CI $[0.016, 0.043]$), and concept supervision raises the AUROC of an 8-unit bottleneck on CTU and TON_IoT, though not on CIC.
3) **Geometry is associated with detectability, not a validated predictor.** Concept-fidelity regularisation has no consistent effect on OOD detection; known-class geometry tracks OOD AUROC on two of three corpora, on the third only through one configuration, and with the tested linear model none predicts detectability consistently on an unseen corpus.
4) **Scorer selection remains open.** The best OOD scorer depends on dataset and architecture, and neither of two known-traffic selection procedures recovers it on every corpus.
5) **Reusable evaluation diagnostics.** Concept replacement, branch agreement, leave-one-configuration-out and leave-

one-corpus-out checks, leak-free feature selection and a source-IP holdout, released with code, splits and models.

Scope. The building blocks, concept bottlenecks [7], sigmoid-relaxed threshold rules [8], [9], per-class Mahalanobis scoring [10] and neural/rule hybridisation, are established, and we claim no new state-of-the-art score. The contribution is an empirical characterisation of where interpretable open-set IoT NIDS hold up and where they do not, with evaluation guidance in the spirit of [4]; we make no streaming, concept-drift or false-alert-rate claims. The findings are stated for the models and the concept and rule vocabularies we authored, and, on CTU-IoT-23, for test traffic from source addresses seen in training (Section IV-A).

II. Related Work

Post-hoc explanation is the dominant answer to NIDS opacity: xNIDS [5] uses sparse group lasso to produce human-readable firewall rules, and GEAD [11] extracts global rules from a trained deep model. The explanation lives outside the model and can stop matching it when the model changes, so we build interpretability into the architecture instead.

CBMs [7] force predictions through human-specified concepts. Concept embedding models [12] extend concepts to higher-dimensional embeddings, and a trained network can be converted into a CBM post hoc [13]. Concept bottlenecks are known to leak: Mahinpei et al. [14] show that concept models carry predictive information their labels do not capture. We reproduce this on IoT flows and relate it to the rule-branch dependence of a neuro-symbolic model and to open-set detectability under one multi-seed, multi-dataset protocol. Closest to our setting, Shoukat et al. [15] apply a concept bottleneck with prototype reasoning to intrusion detection in SDN-based industrial networks; we differ in evaluating open-set detection on IoT flows, comparing against a neuro-symbolic rule model, and measuring decision fidelity and operating-threshold behaviour.

Petersen et al. [8] train binary logic-gate networks through smooth relaxations, and Yang et al. [9] inject logical constraints with the straight-through estimator [16]. NeSy-NIDS applies this line of work to IoT flows, where rules are threshold conjunctions over flow features that an analyst can read and override.

The Mahalanobis detector [10] fits class-conditional Gaussians to learned features, and Fort et al. [17] found that feature quality matters more for OOD performance than the scoring rule. Softmax-based alternatives include maximum softmax probability [18] and the energy score [19]. Vaze et al. [20] report that on visual-recognition benchmarks a better closed-set classifier tends to give better open-set recognition; we test whether this empirical relationship transfers to IoT flows. In security, OWAD [6] detects normality shift and CADE [21] isolates drifting samples, both at the system level, whereas we score individual flows. N-BaIoT [2] and Kitsune [1] established per-device anomaly detection for IoT but neither classifies attacks nor explains its decisions. Arp et al. [4] and Apruzzese et al. [22] identify closed-world evaluation and opacity as obstacles to deployment, and De Keersmaeker et al. [23] survey the public IoT security datasets.

TABLE I: Data and evaluation protocol. Split: stratified 70/15/15 per known class, random at the flow level, fixed split seed.

	CTU-IoT-23	CIC-IoT-2023	TON_IoT
Known classes	4	5	5
Held-out families	9	29	5
Features	31 (raw)	39 (z-scored)	23 (18 z-scored)
Known flows per class	23,000	50,000	50,000
Train / val / test (known)	64,400 / 13,800 / 13,800	175k / 37.5k / 37.5k	175k / 37.5k / 37.5k
Unknown pool	32,344	40.2M	373,857
Unknown draw (all models)	10,000 random	$\leq$2,000/family	$\leq$2,000/family
AUROC	5,000 known test vs. 5,000 flows of the unknown draw		
TPR@5%FPR	threshold on all known test flows; TPR on the whole unknown draw		
Seeds	0–4 (initialisation and batch order only)		

III. Methodology

Fig. 1 shows the evaluation framework, and Algorithms 1 and 2 summarise their training and scoring.

A. Datasets and Protocol

CTU-IoT-23 [24] provides labelled flows from IoT devices captured in a laboratory. Each Zeek conn-log record is joined to per-flow statistics extracted from the packet captures (unmatched records keep them at 0, 1.7% of flows; PartOfAHorizontalPortScan labels are excluded, Supplementary Section SIX); we use 31 of the resulting features (28 numeric, 3 categorical), selected on known-class training flows only. **CIC-IoT-2023** [25] covers 33 attack scenarios across 105 devices; we use 39 of its flow features. **TON_IoT** [26] is an independent testbed whose network subset is derived from Zeek logs; we use 23 features from its connection and protocol logs. TON_IoT enters the CBM scorer and concept-space analyses (Sections IV-B, IV-D) as a third corpus. Its flow logs carry no per-flag counts or inter-arrival times, so three CTU concepts that need them are replaced by connection-state, responsiveness and web-service concepts.

Table I summarises the protocol. Known classes are class-balanced by random sampling, and every held-out family is excluded from training and validation. The split and the OOD evaluation subset are the same for all seeds.

The known sets are: CTU {Benign, C&C-HeartBeat, DDoS, Okiru}; CIC {Benign, DDoS-SYN_Flood, Mirai-greeth_flood, Recon-PortScan, VulnerabilityScan}; TON {normal, scanning, DDoS, password, injection}. The held-out families are, for CTU, C&C, Attack, C&C-HeartBeat-Attack, C&C-FileDownload, C&C-Torii, C&C-Mirai, C&C-HeartBeat-FileDownload, FileDownload and Okiru-Attack; for CIC, the dataset's other 29 attack classes (DDoS, DoS, Mirai, reconnaissance, spoofing, web-attack, brute-force and backdoor classes); and for TON, DoS, XSS, backdoor, ransomware and MITM. Each known set is a benign baseline plus one representative family from each dominant malicious category in the dataset (e.g. volumetric DDoS, botnet C&C or Mirai-family traffic, and scanning where the dataset has it; TON adds credential and web attacks), the traffic an operator could plausibly label early in a deployment. Most held-out families are sub-families and protocol variants of these categories, which is the operational open-set case.

Preprocessing. CIC and TON numeric features are standardised with a scaler fit on the training split. CTU features enter

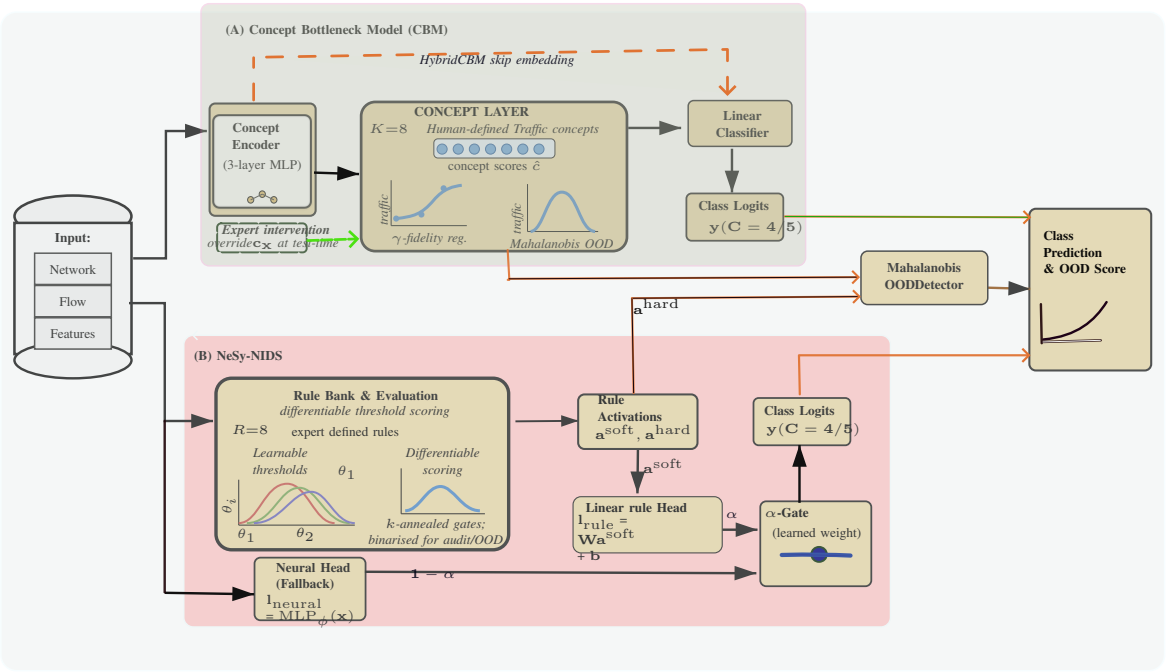


Fig. 1: The Evaluation framework. (A) CBM: concept encoder, concept layer, linear classifier; Mahalanobis OOD scoring in concept space. (B) NeSy-NIDS: rule bank with learnable thresholds and a parallel neural fallback, blended by the learned rule-head weight α. The classifier uses the soft rule activations $\mathbf{a}^{\text{soft}}$; the binarised rule vector $\mathbf{a}^{\text{hard}}$ (Eq. 5) feeds OOD scoring and audit only.

in raw units (missing values set to -1), with categorical fields integer-encoded from vocabularies built on known-class flows. Concept and rule thresholds were written from documented attack signatures and known-class statistics. No preprocessing statistic or concept or rule threshold uses a held-out family. The submitted CTU list (37 features) was an exception: it was selected by random-forest importance on a sample that included every CTU label, the held-out families among them, which may favour CTU OOD separability (CIC and TON use fixed lists). All CTU results here therefore use the same selection re-run on known-class training rows only (31 features; NeSy-NIDS adds back the two counters its expert rules read), and the submitted list is kept as an ablation. The re-selection leaves the accuracy and open-set findings in place, and it raises rather than lowers the AUROC of the configurations most sensitive to the feature list (Table III, AUROC[§]). Every split and evaluation subset is drawn deterministically by the scripts we release.

Robustness splits. We repeat the CBM evaluation on an alternative known set per dataset (CTU {Benign, DDoS, Okiru, C&C}, promoting C&C to known and demoting C&C-HeartBeat to unknown; CIC {Benign, DDoS-ICMP_Flood, Mirai-udpplain, Recon-PortScan, VulnerabilityScan}) and on a timestamp-ordered CTU split, in which each known class is ordered by capture time and the earliest 70% trains and the latest 15% tests. For CTU, the feature list is re-selected by the same random-forest procedure on each experiment's own known-class training rows, so the family it holds out (and any test-period flow) never informs its features. The alternative split draws its own class-balanced sample, since its known classes differ, while the temporal split reorders the main split's sample by time; a control that uses the main split's 31-feature list on the same sample separates the effect of re-selection from that of the sample, and the accuracy and open-set conclusions hold under both.

B. Concept Bottleneck Models

A CBM [7] factors the classifier into two stages,

$$\hat{\mathbf{c}} = g(\mathbf{x}), \quad \hat{y} = h(\hat{\mathbf{c}}), \tag{1}$$

where $g : \mathbb{R}^d \to [0,1]^K$ is the concept encoder (an MLP $d{\to}256{\to}256{\to}64{\to}K$ with LayerNorm and ReLU) and $h : [0,1]^K \to \mathbb{R}^C$ is a linear classifier over the C known classes. We define $K{=}8$ concepts per dataset (e.g. *is_high_rate*: Rate$>$1000 pps; *is_beaconing*: 1.5$<$IAT$<$5 s $\wedge$ 0.1$<$Rate$<$5; *is_incomplete_handshake*: syn_count$>$2 $\wedge$ fin_count$=$0), each a binary label produced by a threshold rule on the raw features (Table II). Because the labels are deterministic functions of the input, *concept accuracy* measures how faithfully the bottleneck reproduces these declared predicates, not whether the predicates are semantically correct. $K{=}8$ balances vocabulary against the sample size the shared-covariance Mahalanobis fit needs.

Training variants. *JointCBM* minimises

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda\, \mathcal{L}_{\text{BCE}} + \frac{\gamma}{K} \|\hat{\mathbf{c}} - \mathbf{c}_{\text{gt}}\|^2, \tag{2}$$

where $\mathbf{c}_{\text{gt}} \in \{0,1\}^K$ are the concept labels, λ weights concept supervision, and $\gamma \in \{0, 0.1, 0.5, 1.0\}$ weights an additional squared-error concept-fidelity term. Both terms push the predicted concepts toward the same labels. *SequentialCBM* trains and freezes g on the concept labels before training h. *HybridCBM* concatenates the 64-dimensional encoder embedding with the K concepts as classifier input, so the classifier

TABLE II: Concept definitions: the threshold rule that produces each binary concept label. CTU thresholds are in raw units; CIC thresholds are z-scores with raw-unit equivalents in parentheses; TON_IoT uses raw Zeek fields and replaces three CTU concepts that need fields its logs lack.

Concept	Definition
CTU-IoT-23 (raw units)	
is_beaconing	1.5<IAT<5.0s ∧ 0.1<Rate<5.0
is_high_rate	Rate >1000 pps
is_syn_heavy	syn_count >8
is_short_connection	0≤duration<0.01s
is_persistent	duration >2.5s
is_asymmetric	resp_pkts=0 ∧ orig_pkts>1
is_single_packet_probe	orig_pkts=1
is_incomplete_handshake	syn_count>2 ∧ fin_count=0
CIC-IoT-2023 (z-scored; raw-unit equivalent in parentheses)	
is_high_rate	rate >0.15 (>2.4×10^4 pps)
is_syn_flood	syn_count >1.5 (>79)
is_udp_dominant	tcp <−1.0 (<0.26)
is_short_connection	iat <−0.00316 (<4.7×10^{-4} s)
is_large_payload	avg >0.4 (>529 B)
is_port_scan	rst_count >1.5 (>4.8)
is_high_variance	variance >0.5 (>8.2×10^5)
is_persistent	fin_count >1.0 (>0.78)
TON_IoT (raw Zeek conn-log units)	
is_high_rate	(src_pkts+dst_pkts)/duration >1000
is_short_connection	0≤duration<0.01s
is_persistent	duration >2.5s
is_asymmetric	dst_pkts=0 ∧ src_pkts>1
is_single_packet_probe	src_pkts=1
is_incomplete_handshake	conn_state ∈ {S0,S1,REJ,RSTOS0,RSTRH,SH,SHR}
is_responsive	dst_pkts >0
is_web_service	service ∈ {http, ssl}

Algorithm 1 CBM Training and OOD Scoring

Require: $\{(\mathbf{x}_i, \mathbf{c}_i, y_i)\}$, λ, γ
Initialise encoder g (d→256→256→64→K, LayerNorm+ReLU), linear classifier h
for each epoch and mini-batch $(\mathbf{X}, \mathbf{C}, \mathbf{Y})$ **do**
  $\hat{\mathbf{C}}=g(\mathbf{X})$; $\mathcal{L}=\mathcal{L}_{\mathrm{CE}}(h(\hat{\mathbf{C}}), \mathbf{Y})+\lambda\mathcal{L}_{\mathrm{BCE}}(\hat{\mathbf{C}}, \mathbf{C})+\frac{\gamma}{K}\|\hat{\mathbf{C}}-\mathbf{C}\|^2$; update g, h
end for
$\boldsymbol{\mu}_c$=mean$\{g(\mathbf{x}_i): y_i=c\}$; $\boldsymbol{\Sigma}$= regularised shared covariance
Inference: $\hat{\mathbf{c}}=g(\mathbf{x})$; $\hat{y}=\arg\max h(\hat{\mathbf{c}})$; $s(\mathbf{x})$ by Eq. 3

can use information outside the concept vocabulary. *Post-hoc CBM* fits a logistic-regression probe per concept to the frozen MLP embedding and a logistic-regression classifier to the probe outputs, both on the training split.

OOD scoring. We apply per-class Mahalanobis scoring [10] in concept space. Class means $\boldsymbol{\mu}_c$ and a shared covariance $\boldsymbol{\Sigma}$ ($\varepsilon=10^{-4}$ ridge, pseudoinverse) are fitted to training concept vectors, and the anomaly score is

$$s(\mathbf{x}) = \min_c (\hat{\mathbf{c}} - \boldsymbol{\mu}_c)^\top \boldsymbol{\Sigma}^{-1} (\hat{\mathbf{c}} - \boldsymbol{\mu}_c). \tag{3}$$

C. NeSy-NIDS: Differentiable Rule Learning

Rule templates. The $R=8$ rule templates per dataset are written from domain knowledge. Each is a conjunction of threshold conditions; representative CTU-IoT-23 rules are `ddos_short_burst` (Rate>1000 ∧ duration<0.1; DDoS), `okiru_low_rate` (Rate<500 ∧ duration<2.5; Okiru) and `benign_single_probe` (orig_pkts<2 ∧ Rate<100; Benign), and several rules start from concept thresholds (e.g. `cc_heartbeat_beaconing` from *is_beaconing*). The full banks with initial thresholds and declared target classes are in Supplementary Table S14. Let $\mathcal{I}_r^{>}$ and $\mathcal{I}_r^{<}$ index the greater-than and less-than conditions of rule r. Following [8], each indicator is replaced by a sigmoid gate and the conjunction by a product:

$$a_r(\mathbf{x}; k) = \prod_{i\in\mathcal{I}_r^{>}} \sigma\big(k(x_i-\theta_i)\big) \cdot \prod_{j\in\mathcal{I}_r^{<}} \sigma\big(k(\theta_j-x_j)\big), \tag{4}$$

so the thresholds θ receive gradients throughout training.

Annealing and binarisation. The steepness k rises linearly from $k_0=1$ to $k_T=10$ over $T_{\mathrm{warm}}=30$ epochs, $k(t) = k_0 + \min(t/T_{\mathrm{warm}}, 1)(k_T-k_0)$. Training and classification use the soft activations of Eq. 4. For audit and OOD scoring, each condition is thresholded and the rule is their logical conjunction,

$$a_r^{\mathrm{hard}} = \prod_{i\in\mathcal{I}_r^{>}} \mathbf{1}[x_i>\theta_i] \prod_{j\in\mathcal{I}_r^{<}} \mathbf{1}[x_j<\theta_j], \tag{5}$$

which is not the same operation as thresholding the soft product.

Architecture and rule-head weight. A linear rule head maps the soft activations, scaled by learned per-rule importance weights, to rule logits $\mathbf{l}_{\mathrm{rule}}$. A parallel MLP (hidden layers of 128 and 64 units, LayerNorm and ReLU) on the raw features produces neural logits $\mathbf{l}_{\mathrm{neural}}$, and

$$\mathbf{l} = \alpha\, \mathbf{l}_{\mathrm{rule}} + (1-\alpha)\, \mathbf{l}_{\mathrm{neural}}, \qquad \alpha=\sigma(w_\alpha), \tag{6}$$

with w_α learned jointly. An optional penalty $\lambda_\alpha(1-\alpha)^2$ pushes α upward. α is a single global mixing weight; because the two heads can emit logits of different magnitudes, its value alone does not measure how much the prediction relies on the rules (Section IV-C).

OOD scoring in rule space. The binary vector $\mathbf{a}^{\mathrm{hard}}(\mathbf{x}) \in \{0,1\}^R$ is scored with Eq. 3, fitted on known-class training activations. Binary vectors are not Gaussian, so this is an approximation; Section IV-B compares it with binary-native and logit-space scorers.

Algorithm 2 NeSy-NIDS Training and OOD Scoring

Require: $\{(\mathbf{x}_i, y_i)\}$, templates with initial thresholds $\boldsymbol{\theta}^{(0)}$, λ_α, k_0=1, k_T=10, T_{warm}
Initialise $\boldsymbol{\theta}=\boldsymbol{\theta}^{(0)}$, rule head $(\mathbf{W}, \mathbf{b}, \mathbf{w}_{\mathrm{imp}})$, MLP ϕ, gate w_α
for epoch $t = 1, \ldots, T$ **do**
  $k=k_0+\min(t/T_{\mathrm{warm}}, 1)(k_T-k_0)$
  for each mini-batch $(\mathbf{X}, \mathbf{Y})$ **do**
    $\mathbf{a}=\mathbf{a}(\mathbf{X}; k)$ (Eq. 4, soft); $\mathbf{l}=\alpha\,\mathbf{l}_{\mathrm{rule}}(\mathbf{a})+(1-\alpha)\,\mathrm{MLP}_\phi(\mathbf{X})$, $\alpha=\sigma(w_\alpha)$
    $\mathcal{L}=\mathcal{L}_{\mathrm{CE}}(\mathbf{l}, \mathbf{Y})+\lambda_\alpha(1-\alpha)^2$; update $\boldsymbol{\theta}, \mathbf{W}, \mathbf{b}, \mathbf{w}_{\mathrm{imp}}, \phi, w_\alpha$
  end for
end for
Fit $\boldsymbol{\mu}_c$, $\boldsymbol{\Sigma}$ on $\mathbf{a}^{\mathrm{hard}}(\mathbf{x}_i)$ (Eq. 5) of the training flows
Inference: classify $\hat{y}=\arg\max \mathbf{l}$ from soft $\mathbf{a}(\mathbf{x}; k_T)$; score $s(\mathbf{x})$ by Eq. 3 on $\mathbf{a}^{\mathrm{hard}}(\mathbf{x})$ (OOD and audit)

D. Open-Set Decision Rule

Both classifiers are closed-set, so we couple the anomaly score to the classifier through a rejection threshold τ:

$$\hat{y}_{\mathrm{open}}(\mathbf{x}) = \begin{cases} \textit{unknown}, & s(\mathbf{x}) > \tau, \\ \arg\max_c\ l_c(\mathbf{x}), & \text{otherwise}, \end{cases} \tag{7}$$

where l_c is the c-th class logit. τ is the 95th percentile of scores on the known-class validation split, so about 5% of known traffic is rejected; no unknown flow is used to set it.

E. Baselines, Metrics and Training

The MLP baseline has the same encoder as the CBM followed by a linear classifier and is scored with Mahalanobis on its 64-dimensional embedding. A decision tree (DT; depth chosen from $\{5, 10, 15, \infty\}$ by validation F1) is paired with per-class Mahalanobis on the raw training features, and a random forest (RF [27]; 100 trees, depth 10) is scored with $1-\max(\hat{p})$. We report weighted F1 on the known test split, and AUROC and TPR at 5% FPR with known test flows as negatives and held-out unknown flows as positives. All hyperparameters and early stopping use the known-class validation split. Adam (lr=10^{-3}, batch 512); CBMs train for up to 50 epochs (patience 10) with λ=0.5, the middle of the grid $\{0.1, 0.5, 1.0\}$: validation F1 differs by at most 0.0005 between λ=0.1 and 0.5, and λ=1.0 is unstable on CTU, where one of five seeds collapses (F1 0.73). Raising λ from 0.1 to 1.0 raises concept accuracy (CTU 0.70→0.94, CIC 0.87→0.98) but moves OOD AUROC in opposite directions (CTU 0.810→0.830, CIC 0.743→0.703; five seeds, Supplementary Table S4); NeSy-NIDS trains for up to 60 epochs (patience 12) with $\lambda_\alpha \in \{0, 0.1\}$.

Evaluation subsets. All models share the training, validation and test splits of their corpus. Every row of Tables III, VI and X, and the NeSy-NIDS λ_α and k_T sweeps, are scored on one fixed set of flows per corpus: 20,000 training flows fit each scorer, AUROC compares 5,000 known test flows with 5,000 flows of the unknown draw (Table I), and TPR sets its threshold at the 95th percentile over all known test flows and is measured on the whole unknown draw; both sets are the same for every model. This TPR is a test-ROC point; the open-set decision uses the validation threshold τ (Section III-D). The scorer × architecture and scorer-selection analyses draw a separate subset per corpus, so a model can differ slightly between tables (CIC MLP 0.807±.007 vs. 0.807±.008; CTU MLP 0.874 vs. 0.879); we compare values only within one analysis.

IV. Results

A. Classification Accuracy

Table III shows NeSy-NIDS and every JointCBM variant tracking the MLP on both datasets. Paired per-seed F1 differences from the MLP have means of at most 0.0001 in magnitude on CTU-IoT-23 (no seed beyond 0.0004) and −0.0004 to +0.0002 on CIC-IoT-2023 (no seed beyond 0.0026). These differences are well inside seed variation; we do not test statistical equivalence. For JointCBM (γ=0) the match also holds per class: every class is within 0.002 of the MLP. The weighted F1 hides a spread across classes: the weakest are CTU Okiru (0.86) and DDoS (0.90), and CIC Recon-PortScan (0.64), VulnerabilityScan (0.70) and Benign (0.76).

The rule-only head reaches only 0.566 on CTU-IoT-23 and 0.750 on CIC-IoT-2023, so the neural fallback covers traffic the eight rule templates do not. SequentialCBM and Post-hoc CBM lose much of their accuracy (CTU 0.567/0.576; CIC 0.732/0.746): consistent with a linear classifier over separately learned concepts being unable to recover the class boundary. The γ variants have near-identical F1 in the main runs; in the supplementary (λ, γ) grid two of the fifteen CTU runs at λ=1.0 collapse (F1 0.73–0.74).

Robustness. On the alternative known sets the MLP reaches F1 0.935 (CTU) and 0.820 (CIC), and every JointCBM variant matches it within 0.001 on both. The timestamp-ordered split lowers four-class weighted F1 from 0.933 to 0.646–0.648 for every model in Table IV except SequentialCBM (0.330). The CTU *DDoS* label spans two capture regimes (a high-rate flood, then a near-zero-rate capture), so the temporal split trains on one and tests on the other, and every model scores F1 0.00 on DDoS on every seed; on the same sample with the main split's 31-feature list DDoS also collapses for every model and seed except two JointCBM γ=0.5 seeds, so the cause is the label's temporal structure, not feature selection. On the three temporally-coherent families the interpretable models match the MLP in aggregate (Coh-3 0.862–0.863) while trading per-class strengths (JointCBM γ=0: Benign 0.841 vs. 0.958, Okiru 0.961 vs. 0.844); only SequentialCBM, which loses Benign, falls behind. The accuracy match is therefore a matched-distribution result.

Source-IP holdout. Random flow splits place flows from the same source addresses in training and test, so we also hold out source IP addresses on two corpora (five seeds; the MLP, the CBM variants and NeSy-NIDS, the last on CTU only). Our merged flow tables keep the source IP but not the capture name, and an address can recur across captures, so a held-out address is not shown to be one physical device: the test separates source addresses. Apart from Benign, each CTU known class comes from 4–7 source addresses, and three of them (192.168.1.198, 192.168.100.111, 192.168.100.108) have enough flows in every known class to form the test set while all other addresses train and validate. On TON_IoT the attack flows of each known class come from a few source addresses, so we hold out 192.168.1.30 or .31 for the attack classes and one address for normal traffic. To test open-set detection on a held-out address, we also score its known flows against its own unknown-family flows (MLP, CBMs and NeSy-NIDS; address .198: mostly C&C; TON_IoT: DoS). On TON_IoT part of the classification performance and AUROC is retained (MLP weighted F1 0.876 on the main split, 0.761 and 0.886 on addresses .30 and .31; DoS AUROC 0.41–0.75 on address .30, 0.44–0.75 on seen addresses), but TPR at 5% FPR is at most 0.7% (.30) and 2.6% (.31), and ≤2.2% on seen addresses. On CTU, whose held-out addresses produce class traffic unlike that of the other addresses (address .198's 3,450 Benign test flows are three distinct feature vectors absent from the Benign training flows), classification fails on all three for every model (mean F1 0.42, 0.14–0.23 and 0.34–0.55, against 0.57–0.93 on the main split), and C&C AUROC falls from 0.82–0.88 to 0.40–0.71. With τ set on validation traffic from the training addresses, the known rejection rate also departs from its 5% target on held-out addresses (up to 50% on address .198, 4–13% on TON_IoT), while unknown recall stays at most 13% (Supplementary Section SV). The CTU results therefore hold for source addresses seen in training; without the capture identifier we did not build a capture-disjoint

split.

B. Open-Set Detection

CTU-IoT-23. Under the leak-free feature selection the highest observed means come from NeSy-NIDS with the α penalty ($0.909\pm.014$) and NeSy-NIDS ($0.904\pm.014$), followed by the MLP ($0.874\pm.002$), SequentialCBM ($0.872\pm.026$) and Post-hoc CBM ($0.871\pm.024$); NeSy-NIDS exceeds the MLP by 0.030 (95% CI $[0.016, 0.043]$; bootstrap over seeds and evaluation flows under these splits, Supplementary Section SVI). The four JointCBM variants reach 0.81–0.84 (seed std 0.04–0.05), HybridCBM 0.822, and the DT and RF baselines 0.679 and 0.657. SequentialCBM, whose concept head is trained on the concept labels alone and frozen, has an observed AUROC similar to the MLP's (difference -0.002, CI $[-0.027, 0.018]$) despite a closed-set F1 of only 0.567. With the originally submitted feature list on the same rows, SequentialCBM and HybridCBM reach only 0.782 and 0.772 and Post-hoc CBM 0.805, so known-class-only selection raises them, while the MLP moves from 0.894 to 0.874 and NeSy-NIDS from 0.885 to 0.904 (Table III, AUROC[§]). NeSy-NIDS also reads the two rule counters that the re-selection drops (Section III-A); given the same 33 columns, the MLP reaches $0.878\pm.001$, JointCBM $\gamma{=}0$ $0.804\pm.021$, SequentialCBM $0.852\pm.038$ and HybridCBM $0.790\pm.069$, so the two counters do not account for NeSy-NIDS's higher mean.

CIC-IoT-2023. The MLP ($0.807\pm.007$) and the DT's raw-feature Mahalanobis (0.802) have the highest AUROC. The JointCBM variants reach 0.699–0.717; SequentialCBM and HybridCBM reach 0.667 and 0.669, and NeSy-NIDS with Mahalanobis only 0.647 (0.672 with the α penalty). Unlike on CTU, SequentialCBM is among the weakest detectors here. The low-dimensional concept and rule spaces separate these unknowns less well than the 64-dimensional MLP embedding and 39 raw features. A dimension-matched control, the JointCBM architecture trained with class labels but without concept supervision ($\lambda{=}\gamma{=}0$, an 8-unit bottleneck; five seeds), reaches $0.719\pm.023$, close to JointCBM (difference $+0.003$, CI $[-0.024, 0.027]$), which is consistent with the CIC gap to the MLP arising mainly from the 8-unit width rather than from the concepts. On CTU the same control is unstable ($0.561\pm.154$; seeds 0.44–0.86, F1 0.867) and concept supervision recovers most of the loss (JointCBM 0.823); on TON_IoT it reaches $0.544\pm.115$ against $0.650\pm.044$ for JointCBM and $0.552\pm.027$ for the MLP.

Concept-fidelity regularisation is not a lever. The sign of $\Delta{=}\gamma_{1.0}{-}\gamma_0$ is opposite on the two main splits, and across five dataset/split combinations it is positive on two and negative on three, with $|\Delta| \leq 0.06$ (Table V); on both main splits the 95% CI of Δ includes zero (CTU $[-0.052, 0.065]$, CIC $[-0.037, 0.002]$), so γ gives no consistent improvement. In a joint (λ, γ) grid (five seeds per cell; Supplementary Section SXVIII) the size of the γ effect also changes with λ: on CTU it grows from -0.004 at $\lambda{=}0.1$ to $+0.030$ at $\lambda{=}1.0$, and on CIC it shrinks from -0.040 to -0.003. Effects this small need multi-seed replication under a fixed scoring protocol.

Which scorer works depends on the representation. Table VI compares per-class Mahalanobis on the NeSy rule vector with two binary-native distances to the nearest known-class centroid (Hamming, Jaccard), the logit-space energy score $-\log \sum_c e^{l_c}$ [19], and a z-normalised Mahalanobis+energy average. On CTU the two rule-vector distances and per-class Mahalanobis all work (0.86–0.90) while energy is below chance (0.46). On CIC no activation-space scorer exceeds 0.65 and only the energy-based scorers detect (0.721–0.743). The Gaussian assumption is therefore not what limits the binary rule space; whether that space separates known from unknown flows at all is. Across the MLP and the six CBM configurations the ranking also depends on architecture (Table VII). On CTU Mahalanobis is better for every architecture, by 0.23–0.49, and also with the submitted feature list. On CIC energy is better for the four JointCBM variants (0.732–0.769 vs. 0.702–0.722), while Mahalanobis stays better for the MLP, SequentialCBM and HybridCBM. On TON_IoT energy is better for all seven (0.640–0.713 vs. 0.550–0.644). No scorer is best across the three datasets and all architectures.

Scorer selection. We tested two procedures for choosing the scorer without the novel traffic. The first holds out one known attack family, retrains JointCBM ($\gamma{=}0$) on the rest, and selects the scorer (Mahalanobis or energy) that better detects the held-out family (five seeds per family; Table VIII). On CTU the family-majority vote recovers the true-best Mahalanobis (two of three families, $9/15$ folds). On CIC it picks Mahalanobis for all four families ($15/20$ folds) although energy is best on the true unknowns: the held-out known families stay separable in activation space while the true unknowns do not, so the proxy is confidently wrong on the corpus where the scorer must switch. The second needs no retraining: it builds synthetic outliers by permuting each feature independently across known validation flows and selects the scorer that better separates them from other validation flows. Against the better scorer on the true unknowns (Table VII), it is right for $25/35$ CTU and $26/35$ CIC models, recovering energy for most CIC JointCBM models, but for only $9/35$ TON_IoT models, where it keeps choosing Mahalanobis; overall ($60/105$) it does no better than always choosing Mahalanobis ($62/105$).

Operating point. Table IX evaluates the deployed $(C{+}1)$-class detector at τ (Section III-D). The known rejection rate stays near its 5% target (4.2–5.8%), and τ has a coefficient of variation across seeds of at most 0.11. Open-set macro-F1 is moderate on CTU and low on CIC, where unknown recall is low (for NeSy-NIDS, seed 0, the scorer rejects about 19% of the pooled unknown flows).

Because unknown flows outnumber known ones about 17:1 in this pool, the unrejected unknowns mainly lower known-class precision, so macro-F1 depends on prevalence. Per-family AUROC ranges widely, and the two architectures catch different families (Supplementary Section SXVI): for JointCBM it runs from near-perfect (CIC DDoS-PSHACK_Flood 0.993, 3,000 flows) to below chance (CIC Mirai-greip_flood 0.15, DDoS-UDP_Fragmentation 0.24), so a strong pooled AUROC does not mean the deployed detector flags every novel family at a usable rate. The equal-family averages of Table IX give CTU families of 2–53 flows the same weight as families of 3,000. At the operating threshold NeSy-NIDS flags fewer than 1% of the flows of 8 of the 29 CIC families, among them

TABLE III: Weighted F1 on the known-class test split; OOD AUROC and TPR@5%FPR (known vs. held-out unknown test flows). F1 is the five-seed mean (seed std $\leq$0.0015; SequentialCBM, Post-hoc CBM and rule-only $\leq$0.0152); AUROC and TPR show mean$\pm$std over five seeds. Rows are grouped as the unconstrained MLP, the CBMs, NeSy-NIDS and the classical DT/RF baselines, whose OOD scores are not computed on a learned representation. **Bold** marks the highest observed mean in each OOD column, not an established ranking. OOD scores are per-class Mahalanobis fitted on training vectors: activations for the CBM, NeSy and MLP rows, concept-probe outputs for Post-hoc CBM, and ‡raw features for DT (this score does not use the tree, so it has no seed variance); †RF uses $1-\max\hat{p}$. All rows share the same training, validation and evaluation flows. CTU-IoT-23 uses the known-class-only feature selection (leak-free; Section III-A); §AUROC with the originally submitted 37-feature CTU list on the same rows (ablation); its F1 differs from the F1 column by at most 0.003.

	CTU-IoT-23				CIC-IoT-2023		
Model	**F1**	**AUROC**	**TPR@5%**	**AUROC§**	**F1**	**AUROC**	**TPR@5%**
MLP Baseline	0.9331	0.874±.002	0.519±.001	0.894±.005	0.8186	**0.807±.007**	0.516±.044
JointCBM (γ=0)	0.9330	0.823±.044	0.376±.199	0.817±.031	0.8183	0.717±.017	0.361±.029
JointCBM (γ=0.1)	0.9331	0.805±.035	0.335±.116	0.821±.033	0.8188	0.700±.009	0.344±.035
JointCBM (γ=0.5)	0.9330	0.844±.036	0.475±.152	0.821±.030	0.8184	0.702±.007	0.333±.031
JointCBM (γ=1.0)	0.9331	0.834±.050	0.520±.088	0.833±.029	0.8182	0.699±.013	0.352±.028
SequentialCBM	0.5673	0.872±.026	**0.550±.049**	0.782±.042	0.7321	0.667±.005	0.177±.015
HybridCBM	0.9331	0.822±.055	0.461±.049	0.772±.073	0.8188	0.669±.015	0.173±.031
Post-hoc CBM	0.5758	0.871±.024	0.462±.100	0.805±.030	0.7457	0.652±.018	0.234±.029
NeSy-NIDS	0.9331	0.904±.014	0.493±.061	0.885±.020	0.8186	0.647±.027	0.217±.056
NeSy + α-reg (λ_α=0.1)	0.9330	**0.909±.014**	0.488±.033	**0.896±.016**	0.8198	0.672±.016	0.187±.051
Rule-only	0.5660	0.870±.011	0.426±.022	0.870±.011	0.7498	0.660±.029	0.156±.036
DT‡	0.9337	0.679±.000	0.251±.000	0.557±.000	0.8145	0.802±.000	**0.549±.000**
RF†	0.9338	0.657±.034	0.446±.003	0.630±.002	0.8057	0.704±.006	0.181±.005

TABLE IV: Temporal (timestamp-ordered) split on CTU-IoT-23, features re-selected on the temporal-train rows (Section III-A): per-class known F1, four-class weighted F1 ($F1_4$), weighted F1 over the three temporally coherent families (Coh-3; DDoS excluded), and OOD AUROC (mean$\pm$std over five seeds). DDoS F1 is 0.00 for every model and seed. $F1_4$ and Coh-3 seed std $\leq$0.001 except SequentialCBM (0.03, 0.04). Each known class has a 3,451-flow test support.

Model	Benign	C&C-HB	DDoS	Okiru	$F1_4$	Coh-3	AUROC
MLP baseline	0.958	0.789	0.00	0.844	0.647	0.863	0.746±.010
JointCBM γ=0	0.841	0.788	0.00	0.961	0.647	0.863	0.758±.062
JointCBM γ=0.1	0.840	0.788	0.00	0.961	0.647	0.863	0.783±.040
JointCBM γ=0.5	0.840	0.788	0.00	0.961	0.647	0.863	0.765±.044
JointCBM γ=1.0	0.880	0.789	0.00	0.922	0.648	0.863	0.764±.050
SequentialCBM	0.104	0.773	0.00	0.445	0.330	0.441	0.839±.012
HybridCBM	0.879	0.787	0.00	0.922	0.647	0.862	0.716±.059
NeSy-NIDS	0.917	0.786	0.00	0.883	0.646	0.862	0.888±.025

TABLE V: JointCBM OOD AUROC vs. γ (mean over five seeds) on every split; $\Delta=\gamma_{1.0}-\gamma_0$.

Split	Data	γ=0	γ=0.1	γ=0.5	γ=1.0	Δ
Standard	CTU	0.823	0.805	0.844	0.834	+0.011
Alternative	CTU	0.837	0.837	0.832	0.809	−0.028
Temporal	CTU	0.758	0.783	0.765	0.764	+0.006
Standard	CIC	0.717	0.700	0.702	0.699	−0.018
Alternative	CIC	0.763	0.765	0.765	0.712	−0.051

TABLE VI: NeSy-NIDS OOD AUROC by scorer (mean over five seeds; std in parentheses). Mahalanobis, Hamming and Jaccard score the binary rule vector; Energy scores the logits; Comb. is their z-normalised average. Bold: highest observed mean per dataset.

Scorer	Space	CTU-IoT-23	CIC-IoT-2023
Mahalanobis (per-class)	activations	**0.904** (.014)	0.647 (.027)
Hamming to centroid	activations	0.858 (.008)	0.640 (.006)
Jaccard to centroid	activations	0.888 (.006)	0.633 (.010)
Energy	logits	0.460 (.077)	0.721 (.032)
Mahalanobis + Energy	both	0.654 (.103)	**0.743** (.020)

TABLE VII: OOD AUROC of per-class Mahalanobis vs. the energy score by architecture (mean over five seeds; std in parentheses). All models of a corpus share one evaluation subset, drawn independently of Table III. Bold: better scorer per architecture and dataset.

	CTU-IoT-23		CIC-IoT-2023		TON_IoT	
Architecture	Mahal.	Energy	Mahal.	Energy	Mahal.	Energy
MLP	**0.879** (.001)	0.386 (.073)	**0.807** (.008)	0.615 (.020)	0.550 (.027)	**0.662** (.051)
JointCBM (γ=0)	**0.830** (.044)	0.476 (.111)	0.722 (.014)	**0.734** (.033)	0.644 (.047)	**0.713** (.051)
JointCBM (γ=0.1)	**0.812** (.034)	0.494 (.136)	0.705 (.006)	**0.732** (.028)	0.640 (.038)	**0.674** (.097)
JointCBM (γ=0.5)	**0.851** (.037)	0.570 (.138)	0.706 (.004)	**0.757** (.014)	0.641 (.032)	**0.694** (.075)
JointCBM (γ=1.0)	**0.842** (.049)	0.612 (.124)	0.702 (.014)	**0.769** (.012)	0.642 (.016)	**0.706** (.075)
SequentialCBM	**0.875** (.026)	0.604 (.085)	**0.665** (.006)	0.636 (.017)	0.603 (.008)	**0.640** (.015)
HybridCBM	**0.827** (.055)	0.365 (.047)	**0.670** (.014)	0.629 (.023)	0.615 (.022)	**0.641** (.032)

XSS (0.6%), SQL injection (0.3%) and backdoor malware (0.6%), and JointCBM 6.5–15.6% of the injection, web and brute-force families. Accurate classification of familiar traffic therefore does not establish dependable detection of unfamiliar attacks, and for these families neither interpretable model offers it. Against five post-hoc detectors on the MLP (five seeds; Table X), the confidence scores (MSP [18], ODIN, energy) sit near or below chance on CTU and are modest on CIC, while the feature-space detectors (deep kNN [28] and Mahalanobis) are strongest; we make no state-of-the-art claim.

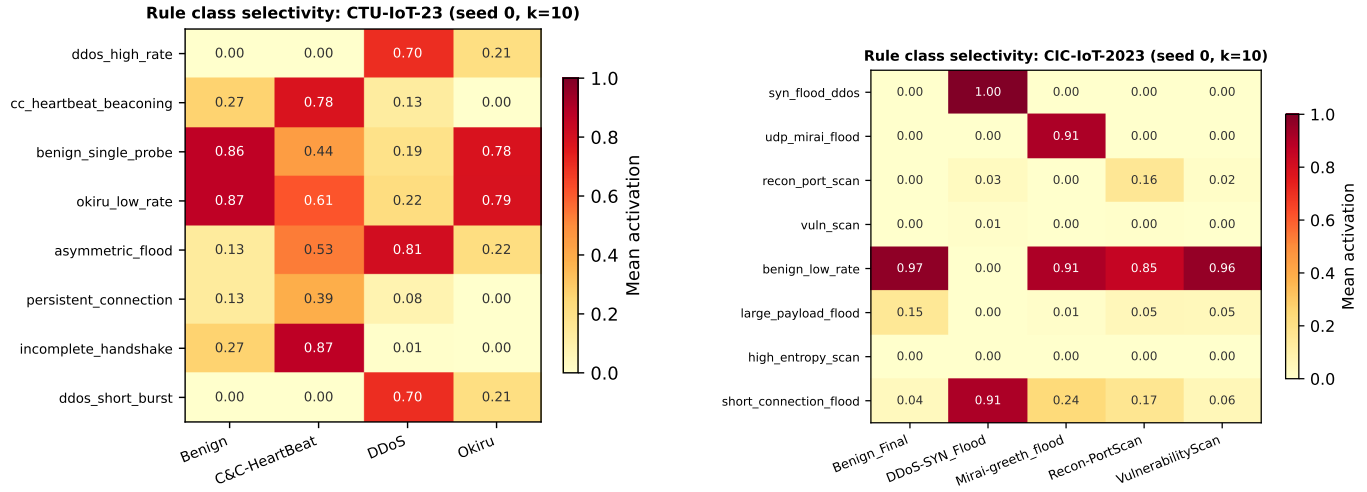


Fig. 2: Mean binarised rule activation (Eq. 5) per known class on the known-class validation split (seed-0 model), CTU-IoT-23 (left) and CIC-IoT-2023 (right). CTU rules fire selectively, leaving within-class variance for the Mahalanobis scorer; CIC rules fire near-deterministically, so there is little to separate on. The pattern is descriptive, not causal (Fig. 4).

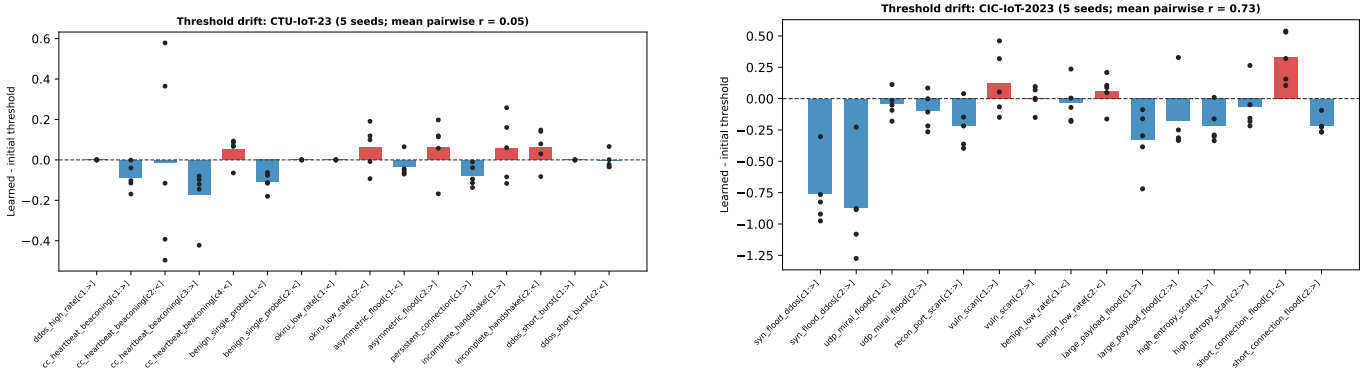


Fig. 3: Per-condition threshold drift (learned − initial) for every NeSy-NIDS rule condition, CTU-IoT-23 (left) and CIC-IoT-2023 (right). Bars are the five-seed mean; dots are individual seeds. Drifts recur across seeds on CIC (mean pairwise r=0.73) but not on CTU (r=0.05), so we read the CIC refinement as systematic and the CTU pattern as indicative only; reproducibility is not semantic validation.

TABLE VIII: Leave-family-out scorer selection (JointCBM γ=0, five seeds per held-out family). AUROCs are five-fold means; "Winner" counts folds selecting Mahalanobis / energy on the held-out family, so it can disagree with the means.

Data	Held-out family	Mahal. AUROC	Energy AUROC	Winner (M/E)
CTU	C&C-HeartBeat	0.90	0.50	5/0
CTU	DDoS	0.86	0.55	4/1
CTU	Okiru	0.22	0.49	0/5
CIC	DDoS-SYN_Flood	0.86	0.54	5/0
CIC	Mirai-greeth_flood	0.85	0.90	4/1
CIC	Recon-PortScan	0.75	0.72	3/2
CIC	VulnerabilityScan	0.70	0.64	3/2

True-best scorer: Mahalanobis (CTU-IoT-23), energy (CIC-IoT-2023).

TABLE IX: Open-set $(C+1)$ evaluation at τ (5% FPR on known validation), mean (std) over five seeds. Known rej.: fraction of known test flows rejected. Macro-F1: over C known classes plus "unknown". Eq.-fam: AUROC and unknown recall at τ averaged over unknown families with equal weight. The JointCBM row is γ=0.

Data	Model	Known rej. (%)	Macro-F1	Eq.-fam AUROC	Eq.-fam recall
CTU	MLP	4.9 (0.2)	0.765 (.002)	0.948	0.88
CTU	JointCBM	5.0 (0.5)	0.654 (.086)	0.818	0.51
CTU	SequentialCBM	5.8 (0.1)	0.489 (.018)	0.746	0.49
CTU	HybridCBM	5.6 (0.1)	0.646 (.018)	0.664	0.35
CTU	NeSy-NIDS	4.8 (0.4)	0.788 (.014)	0.962	0.83
CIC	MLP	5.2 (0.1)	0.267 (.019)	0.806	0.51
CIC	JointCBM	5.2 (0.2)	0.198 (.012)	0.722	0.36
CIC	SequentialCBM	5.0 (0.1)	0.135 (.004)	0.673	0.16
CIC	HybridCBM	5.2 (0.2)	0.131 (.008)	0.676	0.16
CIC	NeSy-NIDS	4.2 (0.6)	0.145 (.017)	0.641	0.20

C. Fidelity of the Concept and Rule Representations

Readable concepts or rules need not account for the predictions made from them, so we test fidelity separately from accuracy (Table XI). **CBM.** We separate the soft predicted concepts, their thresholded values, and the declared predicates used as labels. A linear probe on the soft predicted concepts gains little over one on their thresholded values on CTU-IoT-23 (except for two γ=1.0 seeds) and more on CIC-IoT-2023,

TABLE X: Post-hoc OOD detectors on the unconstrained MLP (AUROC, mean±std over five seeds, on the evaluation flows of Table III).

Dataset	MSP	ODIN	Energy	kNN	Mahal.
CTU-IoT-23	.302±.068	.364±.067	.377±.072	**.879±.003**	.874±.002
CIC-IoT-2023	.609±.012	.698±.013	.607±.019	**.828±.007**	.807±.007

TABLE XI: Fidelity summary (five seeds). CBM ranges span $\gamma \in \{0, 0.1, 0.5, 1.0\}$; NeSy ranges span $\lambda_\alpha \in \{0, \ldots, 1.0\}$. NeSy agreement rows concern the known-class prediction before rejection; the rejection score is computed on the hard rule vector.

	CTU-IoT-23	CIC-IoT-2023
JointCBM		
F1, predicted concepts	0.933	0.818–0.819
F1, declared predicates through own classifier	0.457–0.492	0.624–0.645
F1, logistic regression on declared predicates	0.575	0.718
Probe accuracy, soft minus thresholded concepts	0.001–0.084	0.041–0.058
NeSy-NIDS		
Rule-head weight α	0.38–0.89	0.32–0.95
OOD AUROC across the λ_α sweep	0.893–0.909	0.647–0.673
Class prediction agrees with neural-only head	100%	97–99%
Class prediction agrees with rule-only head	32–51%	45–55%
Hard conjunction agrees with thresholded soft product	>99.9%	99.7–99.9%
Soft activations within 0.1 of 0 or 1	0.87–0.96	0.75–0.80

where the gain grows rather than shrinks with γ (Table XI). A small gain does not establish fidelity, because thresholded predictions can disagree with the declared predicates and still carry class information. We therefore also pass the declared predicates through the model's own classifier, and train a logistic regression on them directly. Both fall far below the learned concepts (on CTU the replacement F1 has seed std 0.02–0.04), so under the evaluated linear classifiers the declared predicates do not reproduce the model's accuracy. Replacement also changes the classifier's input distribution, so these drops test decision fidelity, not a decomposition of accuracy by source. Raising γ improves concept-label agreement on CTU (87.4% to 94.2%; Supplementary Section SII) but leaves the gap within seed variability.

NeSy-NIDS. The hard conjunction almost always agrees with thresholding the soft product, most soft activations are already near 0 or 1 before the cut (known test split), and classifying from the hard activations changes 0.0% (CTU) and 0.9% (CIC) of known test predictions at λ_α=0 (0.0% and up to 3.2% across the sweep), so binarisation has little effect on the blended predictions. At λ_α=0 the deployed known-class prediction (before rejection) agrees with the neural-only head on 100% and 99% of flows and with the rule-only head on only 32% and 45% (seed std 17 and 9 points); raising λ_α over $\{0, 0.05, 0.1, 0.2, 0.5, 1.0\}$ increases α at essentially constant known-class validation F1 (about 0.933 CTU, 0.818–0.819 CIC), and rule agreement rises only modestly with it (CTU 32% to 51%, CIC 45–55%), staying well below the neural agreement (Table XI). OOD AUROC stays within 0.016 on CTU and rises by 0.01–0.03 on CIC as α grows; we do not attribute that rise to greater rule reliance. Branch agreement concerns the class label only; the rejection score is computed on the hard rule vector (Eqs. 3–5). The rules fire in two regimes (Fig. 2):

TABLE XII: Per-rule quality of the NeSy-NIDS rule bank on the known test split (mean over five seeds). Cov.: fraction of flows activating the rule. Prec.: fraction of activating flows in the rule's dominant class. Target prec.: fraction in the rule's declared target class (Table S14), seed std in parentheses. Init.: target precision at the expert initial thresholds, before training. Thr.: relative std of the learned threshold across seeds.

Rule	Cov.	Dominant	Prec.	Target	Target prec.	Init.	Thr.
CTU-IoT-23							
ddos_high_rate	0.23	DDoS	0.76	DDoS	0.76 (0.00)	0.76	0.00
cc_heartbeat_beaconing	0.30	C&C-HB	0.66	C&C-HB	0.66 (0.00)	0.65	0.06
benign_single_probe	0.58	Benign	0.38	Benign	0.38 (0.00)	0.38	0.00
okiru_low_rate	0.63	Benign	0.35	Okiru	0.31 (0.00)	0.31	0.00
asymmetric_flood	0.52	DDoS	0.44	DDoS	0.43 (0.10)	0.48	0.09
persistent_connection	0.15	C&C-HB	0.63	C&C-HB	0.63 (0.00)	0.63	0.02
incomplete_handshake	0.38	C&C-HB	0.63	C&C-HB	0.63 (0.16)	0.76	0.07
ddos_short_burst	0.23	DDoS	0.76	DDoS	0.76 (0.00)	0.76	0.00
CIC-IoT-2023							
syn_flood_ddos	0.20	DDoS-SYN	1.00	DDoS-SYN	1.00 (0.00)	1.00	0.48
udp_mirai_flood	0.09	Mirai	0.85	Mirai	0.85 (0.14)	0.76	0.20
recon_port_scan	0.04	Recon	0.74	Recon	0.74 (0.00)	0.74	0.12
vuln_scan	0.00	DDoS-SYN	0.67	VulnScan	0.01 (0.00)	0.01	0.28
benign_low_rate	0.70	Mirai	0.28	Benign	0.28 (0.02)	0.26	2.90
large_payload_flood	0.06	Benign	0.59	Benign	0.59 (0.06)	0.67	1.91
high_entropy_scan	0.00	Recon	0.46	Recon	0.46 (0.05)	0.44	0.42
short_connection_flood	0.29	DDoS-SYN	0.63	DDoS-SYN	0.63 (0.09)	0.80	0.55

selectively on CTU-IoT-23, leaving within-class variance for the Mahalanobis scorer, but near-deterministically on CIC-IoT-2023, which coincides with the lower CIC AUROC. The learned thresholds drift reproducibly across seeds on CIC (mean pairwise correlation of the per-condition drifts 0.73) but not on CTU (0.05; Fig. 3); reproducibility is not semantic validation. Rule quality is also uneven (Table XII): precision against each rule's declared target class shows the name often mismatches behaviour (`okiru_low_rate` fires mostly on Benign, `vuln_scan` almost never on VulnerabilityScan, and `benign_low_rate` as often on Mirai as on Benign), and several CIC thresholds vary strongly across seeds. These weak rules were already weak at their expert initial thresholds (target precision 0.31, 0.01 and 0.26 before training), so the weakness lies in the authored rules rather than in learning; training mostly widens CIC coverage, sometimes at a cost in precision (`short_connection_flood` 0.80→0.63). In both architectures the inspectable layer records concept or rule activation faithfully but does not by itself explain the classification.

D. Known-Class Geometry: Consistent on Two Corpora, Configuration-Driven on CTU

Across the ten NeSy-NIDS models, mean within-class rule-activation variance tracks OOD AUROC (r=0.973), but the models form two dataset clusters, so this correlation cannot separate a property of the representation from a property of the dataset. We therefore test known-class statistics across the thirty CBM concept-space models per dataset (six configurations × five seeds) on three corpora. On TON_IoT the JointCBM variants reach known-class F1 0.879–0.898 (MLP 0.876) and concept-space AUROC 0.65, above the MLP embedding's 0.55.

Within-class variance. On CIC-IoT-2023 and TON_IoT more within-class variance goes with worse detection ($r = -0.77$ and -0.46). On CTU-IoT-23 the slope is weakly *positive* (+0.24) and comes from SequentialCBM, which shares the

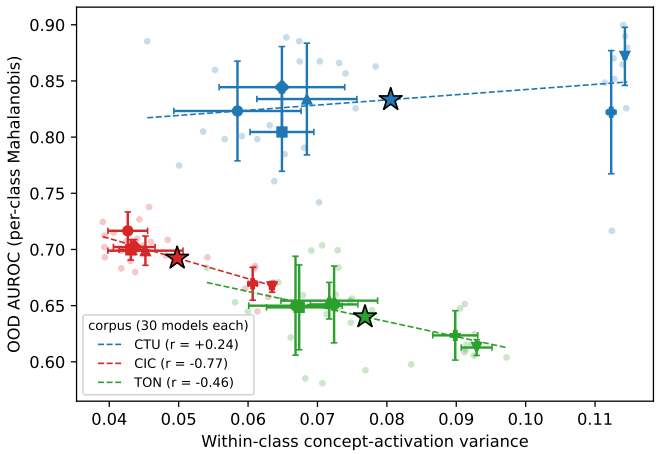


Fig. 4: Within-class concept-activation variance vs. OOD AUROC for the CBM concept-space models, on leak-free feature lists (CTU: known-class-only selection, Section SIX; CIC/TON: fixed). Colour: dataset; marker: configuration; faint points: seeds; solid points: configuration means (± 1 seed std); dashed: per-dataset fits; stars: dataset means. Slopes are negative on CIC and TON; on CTU the weakly positive slope is set by SequentialCBM (top right), and the dataset means are non-monotone.

highest within-class variance with HybridCBM but is also the strongest CTU concept-space detector (0.872, against 0.822 for HybridCBM; Fig. 4). Without SequentialCBM it is $+0.01$ (Table XIII). The within-dataset direction also does not carry across corpora: the three dataset means are non-monotone ($r = +0.36$ over three points), so a corpus's mean known-class variance does not rank its detectability.

Seven statistics. We tested seven statistics computed from known classes only: within-class variance, Fisher trace ratio $\mathrm{tr}(S_B)/\mathrm{tr}(S_W)$, ridge-LDA separability $\mathrm{tr}((S_W+\varepsilon I)^{-1}S_B)$, silhouette, nearest-centroid margin, concept-activation entropy, and closed-set F1. On CIC-IoT-2023 and TON_IoT each keeps the same sign (Table XIII): more known-class separability, or less within-class variance, goes with higher OOD AUROC, as intuition and Vaze et al. [20] predict. On CTU-IoT-23 all seven take the opposite sign ($|r| \leq 0.37$). Without SequentialCBM, which pairs the lowest closed-set F1 and the least separable concept space with one of the highest CTU AUROCs, they shrink to $|r| \leq 0.22$ and three change sign (Table XIII, CTU*), so the CTU reversal is a property of one configuration rather than of the corpus. Correlations pooled over configurations therefore need such a leave-one-configuration-out check before they are read as a property of a dataset. These are within-corpus associations, and they do not transfer consistently: fitting AUROC on each statistic over two corpora and predicting the held-out third, the prediction beats the training-corpora mean AUROC in only 3 of 21 corpus–statistic cases, ranks the held-out models in the right direction in 7 of 21, and identifies the best configuration in 1 of 21.

A manipulation and an interpretation. Retraining NeSy-NIDS at annealing steepness $k_T \in \{2, 5, 10, 20\}$ (five seeds, constant known-class F1) raises within-class variance on CIC ($0.024 \rightarrow 0.037$) but AUROC follows only weakly (r=0.44, 0.61 at $k_T \leq 5$ vs. 0.65 at $k_T \geq 10$); on CTU variance barely moves (0.125–0.128) and is not positively associated with AUROC ($r = -0.15$). This is consistent with a known-class-conditional scorer depending on where the unknown flows fall, not on the known classes alone; we state the result only for the settings tested.

TABLE XIII: Within-dataset Pearson r between each known-class statistic and OOD AUROC over 30 CBM concept-space models per dataset (six configurations $\times$ five seeds). All corpora use feature selection blind to the held-out families (CTU: known-class-only selection, Section SIX; CIC/TON: fixed lists). CIC and TON share every sign; the CTU signs are opposite and fall to $|r| \leq 0.22$ without SequentialCBM (CTU*).

Known-class statistic	CTU	CTU*	CIC	TON
Within-class variance	+0.24	+0.01	−0.77	−0.46
Fisher ratio	−0.18	+0.02	+0.69	+0.47
Ridge-LDA separability	−0.26	−0.08	+0.46	+0.52
Silhouette	−0.22	+0.01	+0.81	+0.51
Nearest-centroid margin	−0.24	−0.02	+0.65	+0.45
Concept-activation entropy	−0.27	−0.06	+0.80	+0.40
Closed-set F1	−0.37	+0.22	+0.53	+0.35
Unknown/known distance ratio†	+0.65	+0.59	+0.56	+0.72

*CTU without SequentialCBM (25 models). †Uses the unknown flows, so it is a proxy of AUROC, not an independent predictor.

V. Discussion

Deployment implications. The threshold τ needs only known validation traffic and held the known rejection rate near its target on the main splits (Section IV-B), though not on held-out source addresses (Section IV-A); neither tested scorer-selection procedure was reliable (limitation (iii)). γ offers no reliable OOD benefit, so effort is better spent matching the scorer to the representation and reading the concept or rule vector within the fidelity limits of Section IV-C.

Computational cost. On CTU-IoT-23, JointCBM adds 296 parameters to the MLP and NeSy-NIDS has 13,313, all under 0.4 MB in `float32`. Per-flow (batch-1) latency on one thread of an Intel Core i9-9820X is 96 μs (MLP), 109 μs (JointCBM) and about 590 μs (NeSy-NIDS, an un-vectorised per-rule loop), measured with the submitted 37-feature input.

Limitations and next steps. (i) *Authored vocabulary.* The concepts and rules were written by us from documented attack signatures [1], [2] and not validated by operators or independent experts; an operator study of the vocabulary is the next step. (ii) *Source-address shift and sampling.* The main splits are random at the flow level, and our holdout separates source IP addresses, not verified physical devices; on TON_IoT, classification performance and AUROC are partly retained under source-IP holdout, but unknown-attack recall remains low and the known-traffic rejection rate can exceed its calibration target, while on CTU the learned models fail to classify held-out addresses (Section IV-A). We built no capture-disjoint split (our merged CTU table lacks the capture identifier), and each split is a single class-balanced sample; corpora with many devices per attack class and multi-sample evaluation are the next steps. (iii) *Scorer selection.* Neither selection rule we tested picks the

better scorer on every corpus (Section IV-B); richer pseudo-unknowns are the next step. (iv) *Geometry as a predictor.* No known-class statistic transfers consistently across corpora with the tested linear model (Section IV-D); more corpora are needed to test any corpus-level property. (v) *Timing* was measured only on a desktop CPU, not on gateway hardware. (vi) *Coverage.* NeSy-NIDS and the temporal, alternative and source-IP splits cover one or two datasets, the geometry three. All corpora are fixed-vector flow representations with hand-authored concepts, so packet-sequence, graph and learned-concept representations remain to be studied.

VI. Conclusion

With our authored concept and rule vocabularies, the interpretable detectors we tested match an unconstrained MLP on known classes from source addresses seen in training, and their representations can aid detection: NeSy-NIDS gives the highest CTU AUROC, and concept supervision improves an 8-unit bottleneck on two of three corpora. Their concepts and rules, however, only partly account for their class predictions, open-set detection depends on dataset, architecture and scorer, and at a usable threshold they miss nearly all flows of several unfamiliar families. Known-class statistics track detectability on two corpora, on the third only through one configuration, and with the tested linear model none predicts it consistently on an unseen corpus. Evaluations of interpretable NIDS should report decision fidelity and per-family recall alongside accuracy, use leak-free feature selection, support open-set claims with multiple seeds, architectures, datasets and held-out source addresses, and check pooled correlations with a leave-one-configuration-out analysis. The next steps are corpora with many devices per class and a scorer-selection rule that holds on every corpus.

References

[1] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," in *Network and Distributed System Security Symposium (NDSS)*, 2018.

[2] Y. Meidan *et al.*, "N-BaIoT: Network-based detection of IoT botnet attacks using deep autoencoders," *IEEE Pervasive Computing*, vol. 17, no. 3, pp. 12–22, 2018.

[3] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.

[4] D. Arp *et al.*, "Dos and don'ts of machine learning in computer security," in *Proceedings of the 31st USENIX Security Symposium*. USENIX Association, 2022, pp. 3971–3988.

[5] F. Wei, H. Li, Z. Zhao, and H. Hu, "xNIDS: Explaining deep learning-based network intrusion detection systems for active intrusion responses," in *Proceedings of the 32nd USENIX Security Symposium*. USENIX Association, 2023, pp. 4337–4354.

[6] D. Han *et al.*, "Anomaly detection in the open world: Normality shift detection, explanation, and adaptation," in *Proceedings of the 2023 Network and Distributed System Security Symposium (NDSS)*. Internet Society, 2023.

[7] P. W. Koh *et al.*, "Concept bottleneck models," in *Proceedings of the 37th International Conference on Machine Learning (ICML)*. PMLR, 2020, pp. 5338–5348.

[8] F. Petersen, C. Borgelt, H. Kuehne, and O. Deussen, "Deep differentiable logic gate networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.

[9] Z. Yang, J. Lee, and C. Park, "Injecting logical constraints into neural networks via straight-through estimators," in *Proceedings of the 39th International Conference on Machine Learning (ICML)*. PMLR, 2022, pp. 25 096–25 122.

[10] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018.

[11] D. Han *et al.*, "Rules refine the riddle: Global explanation for deep learning-based anomaly detection in security applications," in *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2024.

[12] M. E. Zarlenga *et al.*, "Concept embedding models: Beyond the accuracy-explainability trade-off," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.

[13] M. Yuksekgonul, M. Wang, and J. Zou, "Post-hoc concept bottleneck models," in *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023.

[14] A. Mahinpei, J. Clark, I. Lage, F. Doshi-Velez, and W. Pan, "Promises and pitfalls of black-box concept learning models," *arXiv preprint arXiv:2106.13314*, 2021.

[15] S. Shoukat, T. Gao, D. Javeed, and M. Adil, "From evidence to decision: Concept–prototype reasoning for intrusion detection in SDN-based industrial networks," *Computer Networks*, vol. 286, p. 112495, 2026.

[16] Y. Bengio, N. Léonard, and A. Courville, "Estimating or propagating gradients through stochastic neurons for conditional computation," *arXiv preprint arXiv:1308.3432*, 2013.

[17] S. Fort, J. Hu, and B. Lakshminarayanan, "Exploring the limits of out-of-distribution detection," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.

[18] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *Proceedings of the 5th International Conference on Learning Representations (ICLR)*, 2017.

[19] W. Liu, X. Wang, J. D. Owens, and Y. Li, "Energy-based out-of-distribution detection," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[20] S. Vaze, K. Han, A. Vedaldi, and A. Zisserman, "Open-set recognition: A good closed-set classifier is all you need?" in *International Conference on Learning Representations (ICLR)*, 2022.

[21] L. Yang *et al.*, "CADE: Detecting and explaining concept drift samples for security applications," in *Proceedings of the 30th USENIX Security Symposium*. USENIX Association, 2021, pp. 3237–3254.

[22] G. Apruzzese *et al.*, "The role of machine learning in cybersecurity," *Digital Threats: Research and Practice*, vol. 4, no. 1, 2023.

[23] F. De Keersmaeker, Y. Cao, G. K. Ndonda, and R. Sadre, "A survey of public IoT datasets for network security research," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 3, pp. 1808–1840, 2023.

[24] A. Parmisano, S. Garcia, and M. J. Erquiaga, "CTU-IoT-23: A labeled dataset with malicious and benign IoT network traffic," *Stratosphere Laboratory Technical Report*, 2020, czech Technical University in Prague.

[25] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, 2023.

[26] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *IEEE Access*, vol. 8, pp. 165 130–165 150, 2020.

[27] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[28] Y. Sun, Y. Ming, X. Zhu, and Y. Li, "Out-of-distribution detection with deep nearest neighbors," in *International Conference on Machine Learning (ICML)*, 2022.

Ngari Crisphine Macharia (Student Member, IEEE) is pursuing the M.Sc. degree at Southern Illinois University Carbondale, USA (B.Tech., GITAM, India, 2024), working on machine learning for network security.

Ning Weng (Senior Member, IEEE) is a Professor of Electrical, Computer and Biomedical Engineering at Southern Illinois University Carbondale (Ph.D., UMass Amherst, 2005), working on deep packet inspection and network security.